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State Operational Resilience and Infrastructure Interdependence

Housing, Health, Essential Services, Domestic Energy and the Circular Economy

Citing Health, Water and Transport: Three Evidence-Based Cases for Systemic Modelling of Infrastructure Dependencies

dr-bosch.github.io/nullification-loops · Oct. 18th, 2026 · OMB-260331-CSD014

SUBMITTED TO

Joint Committee on European Union Affairs (JCEUA) — Cathaoirleach Mr Barry Ward TD and members; the Joint Committee on Justice, Home Affairs and Migration (JCHAM) — Cathaoirleach Mr Matt Carthy TD and members; the Minister for Housing, Local Government and Heritage; the Minister for Transport and Energy; the Joint Committee on Housing, Local Government and Heritage; and the Office of the Ombudsman (OMB-260331-CSD014), with the Tairiscint Meinhart.

EXECUTIVE SUMMARY

The State operates water supply, health services, transport infrastructure and essential public services through distinct Departments, regulators and statutory bodies. No body is required by statute or convention to model how these services interact under stress. The record does not disclose that any body does so in practice.

Three domains —water infrastructure, health system capital delivery, and transport— now furnish concrete, documented evidence of what that absence produces: a hosepipe ban during a housing emergency, a hospital that reached €2.5 billion from an original estimate of €650 million without a stage-gate that bound any body, and a railway Traffic Management System written down by €50 million after testing revealed the software was not ready for service.

Operations Research is the established discipline for modelling the interactions these cases expose. This paper applies it. The question it puts to the Committee, and to the relevant Ministers, is a simple one: which office holds the function of modelling these dependencies, and where does the analytical capacity sit? If the answer is nowhere, that finding is itself the answer, and it can be recorded as such in the National Strategic Risk Assessment.

Section 1

The Central Question

One question, answerable by an identifiable body.

> *Where in the State does the capacity sit to model how water supply, health capacity, transport infrastructure and essential services interact under stress —and which office assembles the competences it needs?*

S.I. No. 559 of 2024 —which transposes the EU Critical Entities Resilience Directive— designates thirteen essential service sectors and requires the national risk assessment to account for cross-sectoral dependencies and cross-border risks (reg. 11(3) and 11(4)(c)). It designates a competent authority for each sector. It prescribes neither the method nor the body that performs the integration.

Three risk assessment exercises currently run in parallel: the National Strategic Risk Assessment (Department of the Taoiseach), the National Risk Assessment for Ireland 2023 (Government Publications), and the National Cyber Risk Assessment. Each of these continues with a different owner, a different cycle, and a different methodology. In optimisation terms these are three partial solutions to a single coupled problem. No master problem assembles them.

If no body holds this function, that answer is itself the most useful output of this paper. A short written confirmation from any Minister named above, or from the Ombudsman's inquiry, would settle the record.

Section 2

Water: the Infrastructure Below the Infrastructure

Uisce Éireann's testimony to the PAC on Oct. 15th, 2026 provides a precise institutional illustration of the structural problem this paper addresses: a utility performing its own functions while the system-level question remains unowned (e.g. how water interacts with housing, energy and population growth).

The supply position

Uisce Éireann told the Public Accounts Committee on Oct. 15th, 2026 that Dublin's water supply sits at 652 megalitres per day against demand of 645 megalitres — a margin of less than 1.1%. Its head of water operations described this as "on a knife edge." The utility operates more than 700 water treatment plants and manages a network of 65,000 km — the largest per head of population in Europe. Leakage nationally was 38% in 2024. It has since been reduced to approximately 36%, against a 2020-to-2024 target that implied a reduction to under 30%; the Commission for Regulation of Utilities imposed a €31 million penalty for missing that target.

1.1% Dublin supply headroom — 652 Ml/day production, 645 Ml/day demand	**36%** National leakage rate, 2024 (down from 49% in 2014)	**€50bn** Estimated full cost to bring the national network to standard over 25 years	**€31m** CRU penalty on Uisce Éireann for missing leakage and billing targets, 2020–2024

The utility told the PAC: *"We can't fix everything everywhere all at once."* Its head of asset management stated that the long-term funding need is "multiples of what we're getting." Against a current capital investment of approximately €1.3

billion per year, reaching compliance will require approximately €2 billion per year for 25 years — a total of around €50 billion.

The summer 2026 disruption

In July 2026, Uisce Éireann issued a six-week hosepipe ban covering six areas nationally, citing a significant increase in domestic water use (around 20% above normal) and sustained pressure on treatment plants, reservoirs and the distribution network. Night-time restrictions were in place on 39 water supplies. Dublin recorded only 34% of its average monthly rainfall in May 2026. The utility stated that its priority was maintaining water for hospitals, nursing homes and businesses.

The EPA's 2026 drinking water report, published July 1st, 2026, identified 35 water treatment plants at risk of supply disruption due to disinfection and other failures, and called on Uisce Éireann to accelerate its remedial action programme.

The housing dependency

The housing programme and water infrastructure are interdependent. Uisce Éireann's Water for Growth Programme commits over €500 million in drinking-water projects across the eastern region over the next ten years, primarily in response to housing delivery requirements. The Shannon Pipeline Project —a 170 km pipeline from the Parteen Basin to Dublin— is the proposed long-term solution to the supply deficit, with a planning application submitted to An Coimisiún Pleanála in Dec. 2025.

Two issues arise from the O.R. perspective. First, the utility's own projections indicate that Dublin will require 34% more water by 2044, driven by population and economic growth. Whether this projection has been modelled against the housing programme targets, the data-centre connection queue, and the Shannon Pipeline timeline — as a single, coupled optimisation — is not disclosed in any document before this paper. Second, 93% of Ireland's data centres are located in the Greater Dublin Area. The same hosepipe ban that threatened hospitals and households was accompanied by political pressure to account for data-centre water use. Whether both demands are modelled against the same supply projection is an open question.

The structural gap

Uisce Éireann is performing its regulatory function. The CRU is performing its regulatory function. The EPA is performing its enforcement function. No body identified in the public record models how water supply interacts, as a single coupled system, with housing delivery targets, data-centre consents, the electricity grid (which water treatment depends on), and the Shannon Pipeline planning process. The thirteen competent authorities under S.I. 559 of 2024 include water and energy as separate sectors. The dependency between them —water treatment plants require electricity; electricity is required to pump through a 170 km pipeline— appears in neither sector's own risk assessment.

> **Open item — Water**
> Which office, if any, models how drinking-water supply interacts with the housing programme, the data-centre connection queue, the Shannon Pipeline planning timeline, and electricity grid capacity — as a single coupled planning problem? If no body holds this function, that gap is reportable under reg. 11(3) of S.I. 559 of 2024.

Section 3

Health: the National Children's Hospital as a Stage-Gate Lesson

From a budget of €650 million in 2014 to a verified cost exceeding €2.5 billion in 2026, still subject to 900-plus outstanding claims, the National Children's Hospital provides the clearest Irish case study in what the absence of stage-gate discipline produces in large capital programmes with distributed decision authority.

The facts of the overrun

Construction on the National Children's Hospital began in July 2016 on the St James's University Hospital campus in Dublin. The project encompasses the hospital itself and two satellite centres at Tallaght and Connolly. The original budget, set in 2014, was approximately €650 million. By Jan. 2026, Cabinet had approved more than €500 million in additional funding, bringing the approved budget to €2.24 billion: an increase of approximately 30% on the previously approved figure and nearly 250% on the original estimate. Outstanding claims from the contractor at that point numbered more than 900 and were estimated at around €500 million, meaning the final cost may not be known for years after the hospital opens. As of Aug. 2026, the hospital risked missing its twentieth completion deadline.

€650m	€2.5bn+	20th Deadline	900+
Original 2014 estimate	Verified cost, Jan. 2026, before 900+ claims settled	Completion deadline at risk as of Aug. 2026	Outstanding contractor claims, potentially years from resolution

The structural diagnosis

No individual failure produces this outcome. The responsible parties —the National Paediatric Health Development Board, the Department of Health, the HSE, and successive Ministers— each operated within the authority they held. The outcome is the product of a design in which no body was responsible for the system-level question: whether the scope was deliverable at the cost stated before commitment was made. This is a standard finding in large capital projects where decision authority is distributed and no stage-gate function holds the master constraint.

Operations Research addresses this class of problem through three distinct methods. Stage-gate analysis, applied before commitment, tests whether the foundational stage is operationally demonstrated before later-stage expenditure is authorised. Real-options analysis values the right to stop: the option to abandon has positive value only if a gate test is performed and its result is binding. Earned-value management, applied during delivery, distinguishes expenditure from progress: money spent against a phase is not progress unless the phase's output works. All three methods require one body to hold the master problem. The record does not disclose that any body held it.

> *"We have a problem with some of these complex projects."*
> *— Minister Richard Bruton, Feb. 2026*

The health-infrastructure dependency

The Children's Hospital case is relevant to this paper not only as an illustration of project-delivery failure but because the hospital is itself an infrastructure dependency. Its opening affects the three existing children's hospitals at Crumlin, Temple Street and Tallaght, which are scheduled to transfer to the new site on its completion. Until it opens, those hospitals continue to operate at capacity on ageing estates. Every month of delay is a month of continued sub-optimal

paediatric capacity, in a health system that simultaneously records nearly 9,500 elderly patients waiting over 24 hours for a hospital bed (summer 2026).

The Survey (Section 22) maps healthcare operations to its own sub-domain. Patient flow, bed management, discharge velocity, and surge capacity are all tractable queueing problems. Whether the State applies queueing analysis to the phased transition between the existing hospitals and the new campus —modelling the dependency chain from completion date through decant logistics to bed availability— is not disclosed in any document before this paper.

> **Open item — Health**
> Which body holds the function of modelling how the National Children's Hospital's opening timeline affects capacity across the three legacy hospitals, and how that transition interacts with system-wide bed availability and A&E throughput? The stage-gate question — was the 2014 scope estimate testable, and was it tested, before commitment — is reserved for the Ombudsman's inquiry and the PAC.

Section 4

Transport: the TMS Write-Down and the Electric Bus Fleet

The €50 million write-down on Iarnród Éireann's Traffic Management System and the documented delays to the electric bus programme share a structural feature: in both cases, decision authority was distributed across at least three bodies, and no body held the master problem.

The Traffic Management System

In May 2026, Iarnród Éireann wrote down approximately €50 million on its Traffic Management System (TMS), a core component of the National Train Control Centre project. The contract dates from 2020. Testing in mid-2025 identified systemic issues; the supplier's own safety assessor found the software not ready for service. Completion is not expected before 2030, a delay of at least five years from the original programme. The contract was subsequently terminated; the supplier disputes responsibility; and the National Transport Authority has commissioned an independent review.

Body	Role in the TMS programme
Iarnród Éireann	Contracting body; signed the 2020 contract
National Transport Authority	Capital expenditure approval authority
Department of Transport	Sponsoring Department; policy owner
Supplier (terminated)	System developer; disputes responsibility

Three bodies each held part of the decision. None held it whole. The O.R. diagnosis is identical to that applied to the Children's Hospital: in a sequentially dependent system, the prospect of delivery is bounded by the weakest stage, and a stage-gate conditional on demonstrated function at operational scale —applied before the next tranche of commitment is authorised— is the instrument that converts that logic into a binding constraint. No finding is made against any body. The example illustrates what distributed competence produces when no body owns the master problem.

The electric bus fleet

The electric bus procurement programme has recorded documented delivery delays and operational teething difficulties. Bus Éireann and Dublin Bus have both reported periods of reduced fleet availability associated with the transition to battery-electric vehicles, including charging infrastructure constraints, range limitations in cold weather, and supplier delivery timelines. The programme's aggregate cost trajectory and its interaction with the wider transport decarbonisation timeline are a further instance of the stage-dependency risk identified in the TMS case.

No specific figures on the electric bus programme are asserted in this paper without a confirmed source. The programme is cited structurally, as an instance of the same distributed-decision pattern.

The structural read

Case	Write-down / overrun	Bodies involved	Stage-gate applied?
National Children's Hospital	€650m → €2.5bn+	NPHDB, DoH, HSE, Minister	Not documented
Iarnród Éireann TMS	~€50m written down; 5+ yr delay	IÉ, NTA, DoT, supplier	Not evidenced
Electric bus fleet	Delays, availability issues	Bus Éireann, Dublin Bus, NTA, DoT	Not publicly disclosed

The pattern is not coincidental. It is structural. Where authority for a capital programme is divided across at least three bodies and no body holds the master constraint, the programme's cost and schedule will be determined not by any body's decision but by the aggregate of decisions that no body made. Operations Research supplies the remedy: a programme office with the function of holding the master constraint, performing stage-gate analysis against demonstrated outcomes, and applying earned-value management throughout. Whether any such function currently exists within the Irish capital delivery architecture is the open question.

> **Open item — Transport**
> Which office, if any, holds a standing function of stage-gate analysis and earned-value management across large capital programmes in transport infrastructure — and is that function applied before commitment to each phase, or only after problems surface?

Section 5

Essential Services: the April 2026 Fuel Disruption

S.I. No. 559 of 2024 designates thirteen essential service sectors and requires cross-sectoral dependency analysis. Three separate risk exercises satisfy three parts of that requirement independently. No master problem reconciles them.

The April 2026 event

Industrial action beginning on April 7th, 2026 at the Whitegate refinery (Cork) and the Galway and Foynes fuel depots placed access to roughly half of national fuel at risk within days. The industry body Fuels for Ireland reported over 600 forecourts dry by April 10th. The National Emergency Coordination Group was convened. Five Departments were engaged simultaneously: Transport, Energy, Justice, Defence and Finance.

Date	Development
April 7th, 2026	Action begins at Whitegate refinery, Galway and Foynes depots
April 8th, 2026	Department of Transport acknowledges correspondence (ref. TTAS-MO-01932-2026)
April 10th, 2026	Over 100 forecourts reported dry; later revised to approximately 600 affected nationally
Subsequent days	NECG convened; five Departments engaged

No listed risk in the NSRA or NRA 2023 had pre-assigned ownership of fuel logistics as a cross-sectoral contingency before April 2026. The disruption illustrated what a single-node failure in an unmodelled dependency network produces: cascading constraints across road freight, food distribution, health supply chains, domestic heating and agriculture, each sector discovering its exposure to the shared bottleneck in real time.

> **Open item — Essential Services**
> Which office assembles the competences of the thirteen designated competent authorities under S.I. 559 of 2024 into a single cross-sectoral dependency model —and does it perform the cross-boundary analysis required by regulation 11(3)?

Section 6

What This Submission Does Not Do

- It does not ask the Government to adopt any method, technology or supplier.
- It does not allege fault or bad faith in any office-holder or Department.
- It does not assert that any submitted material was controlled, classified or subject to restriction.
- It does not assert that the work submitted would have changed any operational outcome.
- It does not assert a legal duty on any Minister beyond what is stated in S.I. 559 of 2024.
- It does not assert an enforceable promise beyond the 2009 undertaking, the characterisation of which is reserved for the Ombudsman's inquiry (OMB-260331-CSD014).
- It takes no position on housing targets, energy policy, water strategy, health service structures or any policy question within the competence of the relevant Ministers.
- It does not name any individual as having acted improperly.
- It does not make causal claims linking any correspondence to any operational outcome.

Section 7

Questions Answerable Yes or No

A written response from any Minister named above, or from the Ombudsman's inquiry, addressing each of these five questions would settle the record.

1. Does the Department of Housing, Local Government and Heritage hold, or have access to, a standing model of how housing delivery interacts with water supply, grid connection, transport and essential service capacity?
2. Does Uisce Éireann's forward planning for the Shannon Pipeline and the Water for Growth Programme incorporate a model of how data-centre water demand and the housing programme jointly affect the Dublin supply position over the decade to 2034?
3. Does the Department of Health hold a standing model of how the National Children's Hospital's opening timeline affects system-wide bed capacity, and was stage-gate analysis applied before each major commitment decision on that programme?
4. Does the Department of Transport hold a standing function of stage-gate and earned-value management for large capital programmes —and does that function apply before commitment to each phase, conditional on demonstrated operational performance?
5. Which office, if any, assembles the competences of the thirteen designated competent authorities under S.I. No. 559 of 2024 into a single cross-sectoral dependency analysis, and does it perform the cross-boundary analysis required by regulation 11(3)?

Why this is restated to the JCHAM and the JCEUA

Both Committees have received this material before. The prior contacts are recorded in the Ombudsman complaint (OMB-260331-CSD014) and in the submission to the European Commission's public consultation on dual-use goods (DG TRADE). In summary, the JCHAM has acknowledged four items of correspondence between September 2025 and March 2026 (R2025 0106, R2025 0221, R2025 0229 and R2026 0068); none records a determination or a referral to a competent body. Two further items, the formal complaint of July 17th, 2025 and the correspondence of November 26th, 2025, are not recorded as acknowledged. That correspondence addressed Operations Research in general terms and the question of remit.

This paper restates the position for three reasons. It puts the question as three documented cases and five questions answerable yes or no, rather than as a question of method or remit. It lets each Committee respond on the facts without first deciding whether Operations Research falls within its remit; a statement that the function is not held, or that it sits with another body, settles the record as fully as an affirmative answer. And it sets the whole position out in one document, so that neither Committee need consult earlier files to see what is being asked. Nothing here is a finding about any office-holder; it records where the matter stands.

Sources

1. Uisce Éireann testimony, Public Accounts Committee, Oct. 15th, 2026 — Irish Examiner ("We can't fix everything everywhere all at once") and Irish Times ("Improving water system to cost €50bn")
2. Commission for Regulation of Utilities, €31m penalty on Uisce Éireann for missing 2020–2024 leakage and billing targets — Irish Times, July 9th, 2026
3. Uisce Éireann six-week hosepipe ban announcement — RTÉ News, July 13th–14th, 2026
4. RTÉ News, "How can a country as wet as Ireland need hosepipe bans?" — July 21st, 2026 (leakage 38%; 85% of Dublin supply from the Liffey; 34% more water required by 2044)
5. EPA, Drinking Water Report 2025 — RTÉ News, July 1st, 2026 (35 treatment plants at risk; 24 supplies with E. coli incidents)
6. The Journal.ie, "The hosepipe ban has led to criticism of Ireland's data centres" — July 2026 (93% of data centres in the Greater Dublin Area)
7. Uisce Éireann, Water for Growth Programme (€500m eastern region investment) — RTÉ News, April 16th, 2026
8. National Children's Hospital — Cabinet approval of €500m additional funding, total to €2.24bn — Leinster Leader, Jan. 29th, 2026; Business Post; Social Democrats statement (€2.5bn+ figure); Irish Times, March 6th, 2026 ("Final cost may not be known for years")
9. National Children's Hospital — 20th deadline at risk — Irish Examiner, Aug. 2026; FrontPages.eu briefing, Aug. 7th, 2026
10. Iarnród Éireann TMS write-down — Irish Times and RTÉ News, May 2026; RTÉ News, June 10th, 2026 and July 18th, 2026; Irish Times, Sept. 8th, 2026 (contract termination); PAC hearing, July 16th, 2026
11. April 2026 fuel disruption — RTÉ News, April 10th and 13th, 2026; TTAS-MO-01932-2026 (Department of Transport acknowledgement, April 8th, 2026)
12. S.I. No. 559 of 2024 (European Union (Resilience of Critical Entities) Regulations 2024), regs 7, 8, 11
13. CSO, Data Centres Metered Electricity Consumption 2024 and 2025
14. Submission to the National Strategic Risk Assessment 2026, Department of the Taoiseach — Oct. 9th, 2026
15. Interregnum Nullificans v20e, Statement of Record — Oct. 16th, 2026
16. Ombudsman complaint OMB-260331-CSD014 (active)
17. Comprehensive Survey of O.R. Sub-Domains and their Theoretical Foundations — submitted Feb. 5th, 2026
18. Submission to the European Commission public consultation on dual-use goods (DG TRADE)
19. JCHAM acknowledgements: R2025 0106 (Sept. 9th, 2025); R2025 0221 (Dec. 15th, 2025); R2025 0229 (Dec. 19th, 2025); R2026 0068 (Mar. 3rd, 2026)
